

Experiment Report: MNIST Digits Experiment (Low-Level)

Metadata

- *Description:* In this experiment, a neural network is trained with different hyperparameters to classify MNIST digits.
- *Start Time:* 2025-03-23 12:37:22
- *Last Resume Time:* 2025-03-23 12:39:11
- *Total Duration:* 0:00:29.47
- *Directory:* [Link](#)

Summary

Hyperparameters

	units1	units2	learning_rate	Chapters
Trial 2	256	128	1.0000e-03	Chapter
Trial 1	128	64	1.0000e-03	Chapter

Test Results

	accuracy	precision	recall	f1	Chapters
Trial 2	0.9882	0.9525	0.9247	0.9384	Chapter
Trial 1	0.9862	0.9502	0.9129	0.9312	Chapter

Trial 2

- *Start Time:* 2025-03-23 12:39:25
- *Duration:* 13.769
- *Directory:* [Link](#)

Hyperparameters:

Hyperparameter	Value
units1	256
units2	128
learning_rate	0.001

Evaluation Metrics:

	train	val	test
accuracy	0.9882	0.9867	0.9882
precision	0.9555	0.9503	0.9525
recall	0.9273	0.9174	0.9247
f1	0.9412	0.9336	0.9384

Figures:

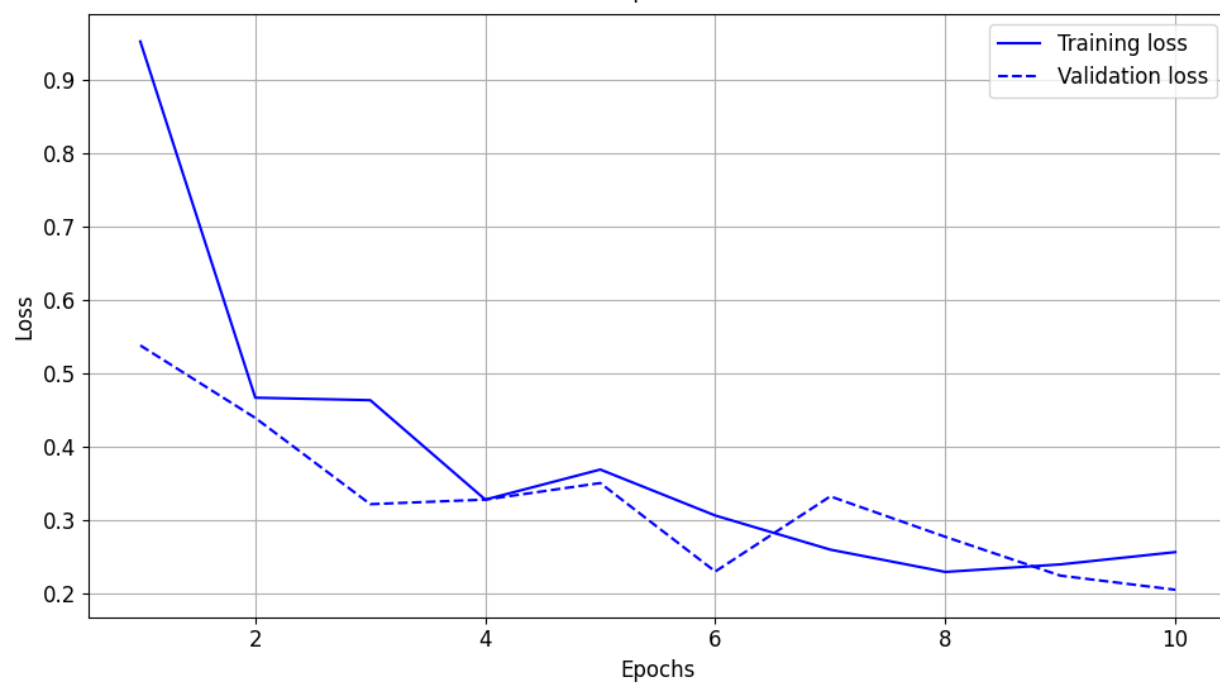
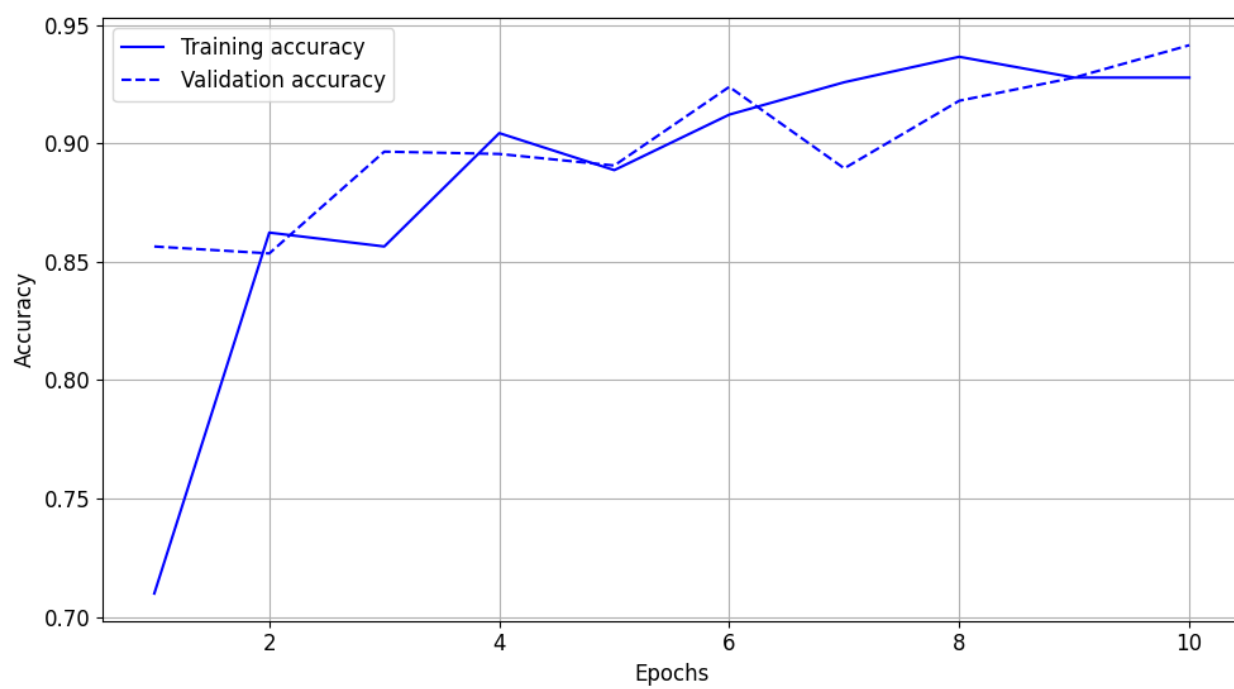
Model Summary

Model: "sequential_1"

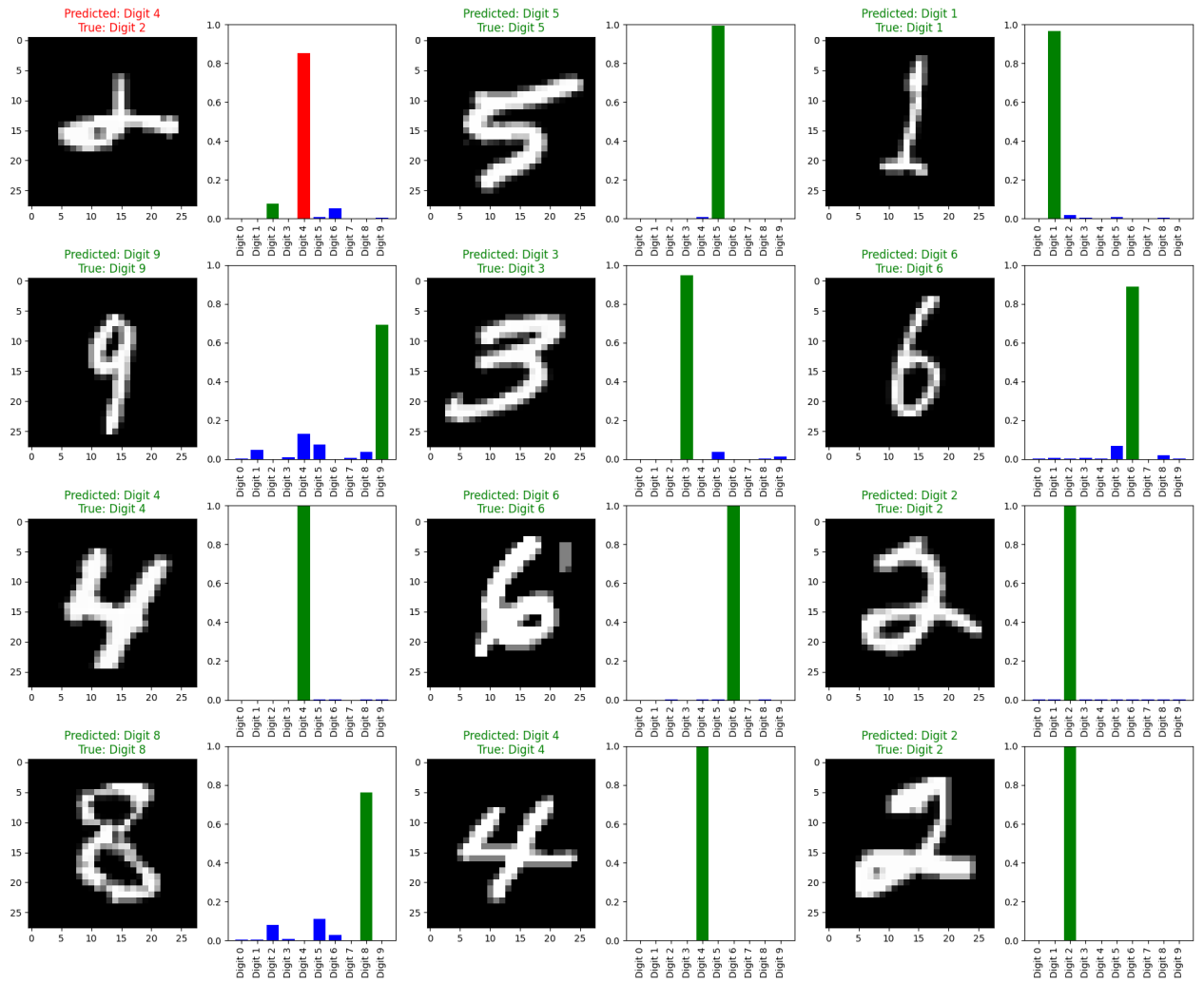
Layer (type)	Output Shape	Param #
flatten_1 (Flatten)	(None, 2352)	0
dense_3 (Dense)	(None, 256)	602,368
dense_4 (Dense)	(None, 128)	32,896
dense_5 (Dense)	(None, 10)	1,290

Total params: 1,909,664 (7.28 MB)
Trainable params: 636,554 (2.43 MB)
Non-trainable params: 0 (0.00 B)
Optimizer params: 1,273,110 (4.86 MB)

Training History



Results



Detailed Report of Test Set:

	precision	recall	f1-score	support
Digit 0	0.9617	0.9784	0.9700	693
Digit 1	0.9553	0.9795	0.9672	829
Digit 2	0.9468	0.9282	0.9374	710
Digit 3	0.9529	0.9046	0.9281	671
Digit 4	0.9495	0.9342	0.9418	684
Digit 5	0.9037	0.9530	0.9277	660
Digit 6	0.9808	0.9472	0.9637	701
Digit 7	0.9426	0.9650	0.9537	715
Digit 8	0.8957	0.9320	0.9135	691
Digit 9	0.9186	0.8761	0.8969	670
micro avg	0.9409	0.9409	0.9409	7024
macro avg	0.9408	0.9398	0.9400	7024
weighted avg	0.9413	0.9409	0.9408	7024
samples avg	0.9409	0.9409	0.9409	7024

Trial 1

- *Start Time:* 2025-03-23 12:39:11
- *Duration:* 13.940
- *Directory:* [Link](#)

Hyperparameters:

Hyperparameter	Value
units1	128
units2	64
learning_rate	0.001

Evaluation Metrics:

	train	val	test
accuracy	0.9857	0.9850	0.9862
precision	0.9473	0.9456	0.9502
recall	0.9121	0.9044	0.9129
f1	0.9294	0.9245	0.9312

Figures:

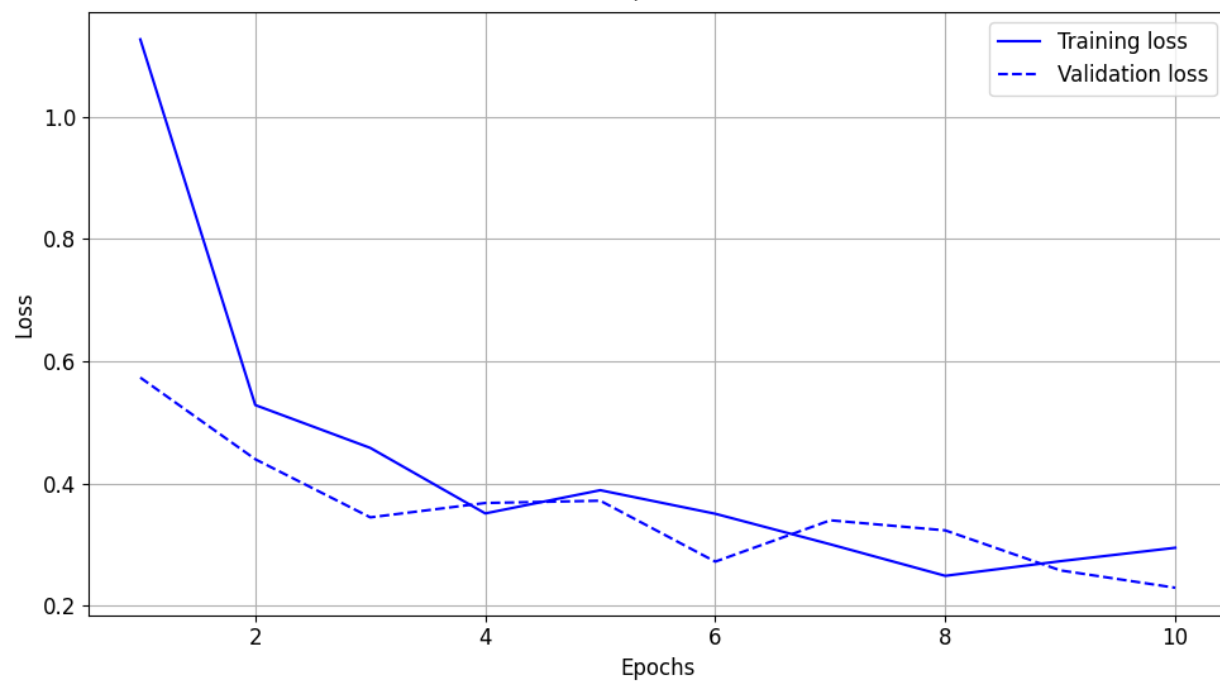
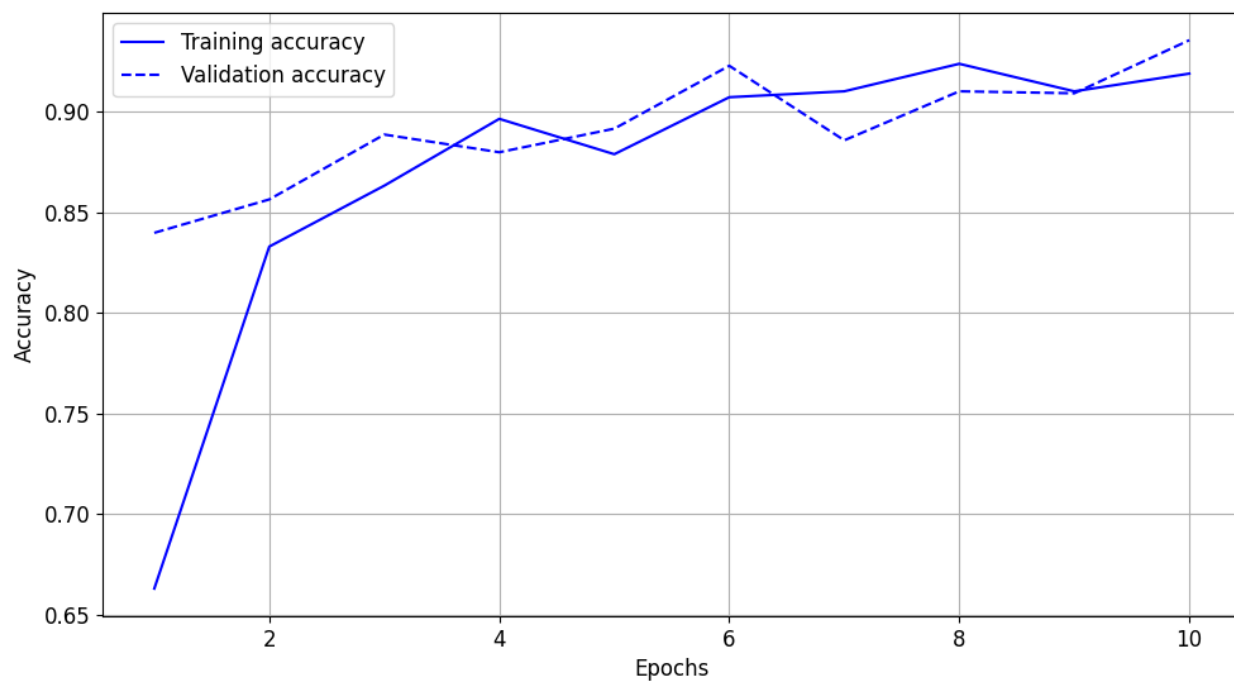
Model Summary

Model: "sequential"

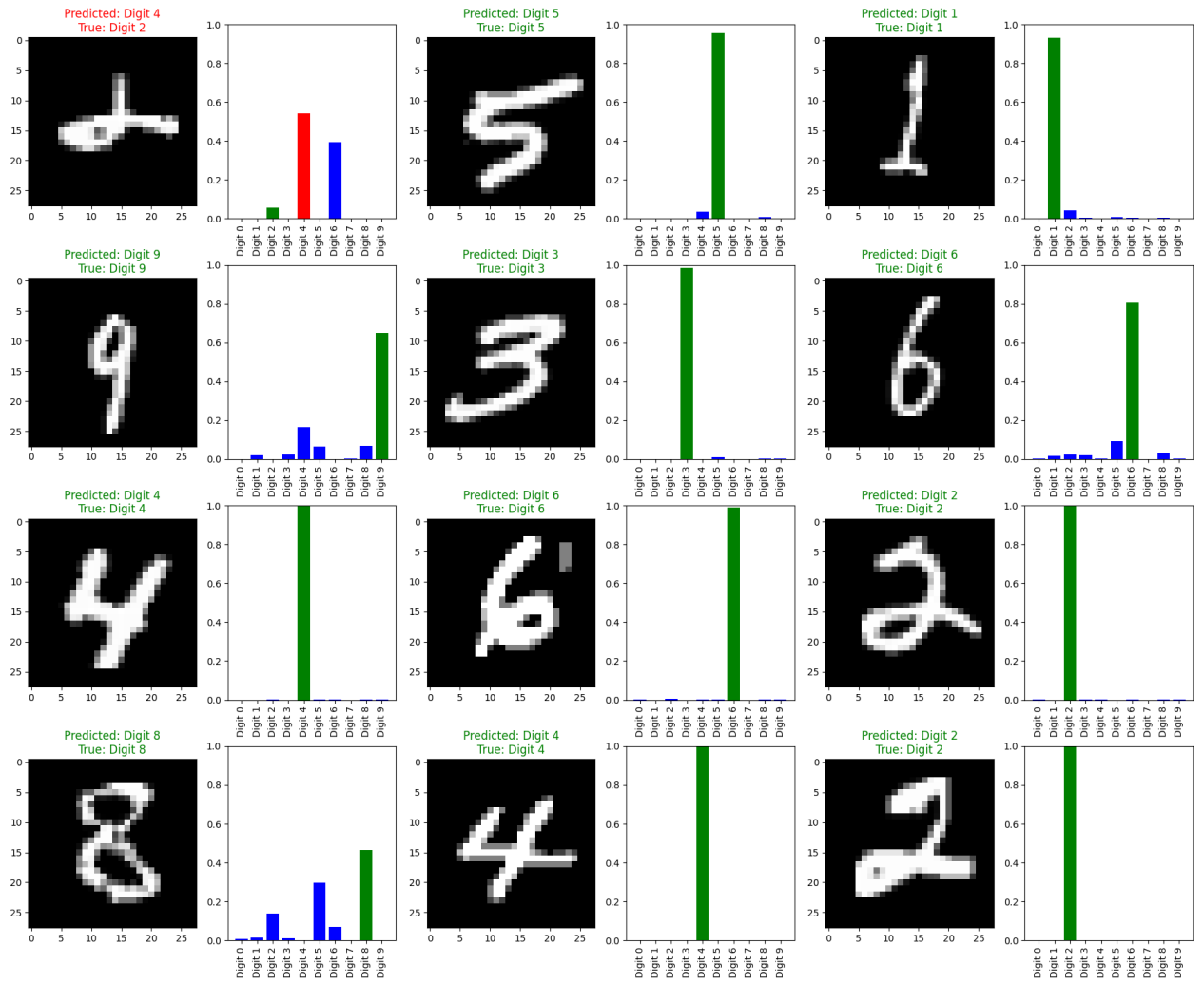
Layer (type)	Output Shape	Param #
flatten (Flatten)	(None, 2352)	0
dense (Dense)	(None, 128)	301,184
dense_1 (Dense)	(None, 64)	8,256
dense_2 (Dense)	(None, 10)	650

Total params: 930,272 (3.55 MB)
Trainable params: 310,090 (1.18 MB)
Non-trainable params: 0 (0.00 B)
Optimizer params: 620,182 (2.37 MB)

Training History



Results



Detailed Report of Test Set:

	precision	recall	f1-score	support
Digit 0	0.9693	0.9582	0.9637	693
Digit 1	0.9397	0.9783	0.9586	829
Digit 2	0.9062	0.9394	0.9225	710
Digit 3	0.9063	0.9225	0.9143	671
Digit 4	0.9503	0.9225	0.9362	684
Digit 5	0.8922	0.9152	0.9035	660
Digit 6	0.9734	0.9387	0.9557	701
Digit 7	0.9371	0.9580	0.9474	715
Digit 8	0.9156	0.9103	0.9129	691
Digit 9	0.9212	0.8552	0.8870	670
micro avg	0.9312	0.9312	0.9312	7024
macro avg	0.9311	0.9298	0.9302	7024
weighted avg	0.9316	0.9312	0.9311	7024
samples avg	0.9312	0.9312	0.9312	7024